CISCO
WEBEX

VIBE
VEL

PROBLEM 2 · VOICE & LANGUAGE INTELLIGENCE LAYER · CASE COMPETITION 2025

VIBE / VEL — Voice & Language Intelligence Layer

A real-time, in-meeting voice and language AI layer that boosts and cleans voice, translates between native and English in both speech and text, simplifies technical jargon for non-tech users, and enables type-to-speak so VIBE speaks on your behalf — all inside Webex. No deepfake risk. Every action auditable.

A · Onboarding   B · Live Speak Path   C · Type-to-Speak
D · Q&A Loop   E · End & Logging   AI Processing   Decision Gate
Data / Output

▶️ START — USER JOINS WEBEX MEETING

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A · USER ENTRY & ONBOARDING

SETUP

⚙ TECH STACK

ASR	Whisper / Cisco AI
NLP	Custom LLM + rules
TTS	Neural voice synthesis
MT	Neural Machine Translate
DSP	DNN noise + EQ
Codec	Webex AI Codec

🌍 LANGUAGES

Hindi · Spanish
Mandarin · French
Arabic · Portuguese
Japanese · Korean
+ 90 more via MT

🔒 PRIVACY & TRUST

1 **User Enables VIBE / VEL in Webex**

Toggle appears in meeting controls bar · Zero configuration required to activate

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2 **Select Active Mode(s)**

One or more modes may be active simultaneously:

🔊 Voice Boost | 🌐 Live Translation | ⌨️ Type-to-Speak | 🧩 Simple Mode (Non-Tech)

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3 **Select Language & Profile**

Native Language: Hindi · Spanish · Mandarin · French · Arabic · +90 more

Tech Level: Technical | Non-Technical

Voice Preset: Clear · Loud · Warm · Broadcast-style

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❓ Is user speaking or typing?

🎙️ SPEAKING → Path B | ⌨️ TYPING → Path C

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B · LIVE AUDIO PATH — WHEN USER SPEAKS

SPEAK PATH

1 🎤 **Microphone Audio Captured**

Raw PCM stream · Per-speaker separation via Webex AI Codec · Real-time buffer pipeline

No deepfake output

Full VIBE label on TTS

Auditable transcript

Zero data retention opt

User-controlled toggle

📊 KPIS

Noise reduction: 95%+

ASR accuracy: 98%

Translation latency: <1s

TTS naturalness: 4.5/5

Type-to-speak: <2s

2 **Audio Intelligence Front-End**

DNN Noise Removal: 150+ noise types (café, keyboard, HVAC, crowd)
Echo Cancellation & Auto Gain Control
Audio tuned to "VIBE mode" — optimized EQ profile per preset
Smart Room Detection: reverb + acoustic fingerprint adjustment

3 **Voice Boost & EQ Engine**

Apply selected preset: Clear · Loud · Warm · Broadcast-style
Adaptive to environment: noisy café ≠ quiet office ≠ conference room
Waveform normalization · Consistent output level to all participants

4 **Speech-to-Text (ASR)**

Real-time transcription in user's native language · Multi-accent support
Per-speaker diarization: who said what, timestamped
Filler word detection: removes "um / uh / like" from clean transcript

5 **NLP Layer — Conditional Processing**

Detect: language · intent · domain · tech level · sentiment

🧩 **Simple Mode ON**

Rewrites jargon

🌐 **Translation ON**

Native language →

📊 **Always Active**

Intent & domain

jargon →
plain
language
Adds real-world analogies
e.g. "API" → "a waiter between kitchen & table"

English
Or to any target language
Preserves technical terms

tagging
Clarity scoring
Real-time caption prep

6 **Text-to-Speech Output — VIBE Voice**

Converts processed/translated text → natural neural speech
Consistent "VIBE persona" voice identity per user across the session
Target: English (global) or specified audience language
Emotional tone matching: excited text is delivered with energy

7 📡 **Delivered to All Participants**

Participants hear: cleaned · boosted · translated VIBE voice
Per-user captions in their own selected language
Each attendee independently controls their own caption language

C · TYPE-TO-SPEAK PATH — WHEN USER CANNOT SPEAK

TYPE PATH

1 ⌨️ **User Switches to Type-to-Speak**

Triggers: muted · too noisy · broken mic · speech difficulty · large-meeting shyness
Auto-suggest: VIBE detects prolonged silence + high ambient noise → offers mode switch

2 **User Types in VIBE Panel or Chat**

Any language · Can paste code, data, tables · Drafts

visible only to user before send

3 **NLP Processing**

Detect language & intent · Simplify or professionalize based on active mode
Translate to meeting language (→ English default)
"Polish mode": convert quick draft into fluent spoken-style language

4 **VIBE Agent Speaks on Behalf of User**

Neural TTS voice speaks typed message aloud in the meeting
On-screen label: "🎙️ **Spoken by VIBE for [UserName]**" — fully transparent, no deception
User selects from 3 VIBE voice personas per session

5 **Caption & Chat Sync**

Caption: "from [UserName] via VIBE ⌨️" · Auto-appended to meeting transcript with source tag
Chat thread logs typed input + spoken output side-by-side

D · Q&A CLOSE-LOOP — NON-NATIVE / NON-TECH USERS

Q&A LOOP

1 🙋 User Asks in Own Language

Voice or Type-to-Speak · Any language
e.g. "API integration kaise kaam karti hai?" (Hindi)

· "¿Cuánto cuesta el plan?" (Spanish)

▼

2 ASR + Translation → English for Presenter

Native speech → English text · Displayed as caption to presenter
Presenter sees: "*[Priya] asked (translated): How does the API integration work?*"

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3 Presenter Answers in English

Presenter responds normally · VIBE captures the answer in real-time
Presenter can tag response: "Simplified answer" to trigger extra NLP treatment

▼

4 Translation + Simplification Back to User

English answer → translated to user's native language
Simple Mode: adds analogy + concrete example
e.g. "Think of an API like a power outlet — your device doesn't need to know how electricity is generated"

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🔊 Audio Output

Answer spoken in native

📝 Text Output

Simplified text +

language
via VIBE voice ·
Private to user

analogy
shown in user's caption pane
Bookmarkable for VIBE Summary

E · END & LOGGING

POST-MEETING

1 VIBE Transcript

Original language layer
English translation layer
Simplified text layer
Timestamps + speaker IDs
Type-to-Speak logs marked ⌨️

2 VIBE Summary Pack

Key Q&A in native language
Simple explanations + examples
Downloadable by each user
Shareable via Webex / email

3 Session Analytics

Voice clarity score trend
Translation accuracy log
Q&A resolution rate
VIBE usage breakdown

🔒 PRIVACY GUARANTEE

Transparency & Trust Layer — Always Active

Every VIBE-synthesized voice is labeled on-screen: "Spoken by VIBE for [User]" — no hidden AI speech
Full audit trail: every VIBE action (boost, translate, TTS) logged with user consent timestamps
Zero data retention option: transcript deleted on meeting end if user opts out of VIBE Summary
VIBE never impersonates: voice persona is distinct from the user's actual recorded voice

■ END — VIBE SESSION COMPLETE

⬡ VIBE / VEL — COMPONENT ARCHITECTURE OVERVIEW

Audio Layer	AI / NLP Layer	Type-to-Speak Layer	Output & Logging
Webex AI Codec DNN Noise Removal Echo Cancellation Auto Gain Control Voice Boost EQ Per-Speaker Separation	ASR (Whisper / Cisco) Neural Machine Translation Jargon Simplification LLM Intent & Domain Detection Neural TTS Synthesis VIBE Persona Voice Model	Text Input Panel (VIBE UI) Language Detection Polish / Simplify Mode TTS Agent Voice Caption + Chat Sync "via VIBE" Transparency Tag	Per-User Caption Stream VIBE Transcript (3 layers) VIBE Summary Download Session Analytics Audit Log (all VIBE actions) Zero-Retention Option